



Text gives you information about a subject.

Hands on

Want to try something for yourself? Then look at a "Hands on" tip.

Properties of matter

Some materials are hard and brittle, while others are flexible. Some materials are colourful, while others are transparent. These kinds of features are called "properties".

Hands on tells you how to get stuck in and try an experiment for yourself.

Buttons contain mini facts: quick information at your fingertips.

- Plasticity** is how well a solid can be reshaped.
- Conductivity** is how well a material lets electricity or heat travel through it.
- Malleability** is how well a solid can be shaped without breaking.
- Tensile strength** is how much a material can stretch without breaking.
- Flammability** is how easily and quickly a substance will catch fire.
- Reflectivity** is how well a material reflects light. Water reflects well.
- Transparency** is how well a material lets light pass through it.
- Flexibility** is how easily a material can be bent.
- Solubility** is how well a substance will dissolve, such as salt in water.

Does it float?
It's easy to learn about some properties, such as the ability to float. The amount of matter in a certain volume of an object is called its density. Objects and liquids float on liquids of a higher density and sink through liquids of a lower density.

A good insulator
Heat cannot easily pass through some materials. These are known as insulators. For example, aerogel can completely block the heat of a flame. But don't try this at home!

Properties of matter

Brittleness
Some materials, such as glass, are very brittle and will break when pushed out of shape. Safety glass is designed to crack rather than break.

Compressibility
Gases can be squashed, or compressed, by squeezing more into the same space. This is what happens when you pump up a tyre.

Hardness
A scientist called Friedrich Mohs created a scale of ten minerals to compare how hard they are. Many materials are graded on this scale.

Hands on
Collect some different pebbles and put them in order of hardness. A pebble is harder than another if it scratches it. This is how Mohs worked out his scale.

A smooth flow
Some liquids flow more easily than others. It depends on their "stickiness", or viscosity. Hot lava from a volcano flows slowly because it is sticky.

Photographs show you information about a subject.

Quick quiz questions are at the bottom of each page.



The universe

Men on the moon

On 20 July 1969, Neil Armstrong became the first person to walk on the surface of the moon. He was joined by Buzz Aldrin. A third astronaut, Mike Collins, remained in orbit with the command and service modules.

weird or what?
The lunar module computer on Apollo 11 had just 71K of memory. Some calculators can now store more than 500K.

What did they do?
Armstrong and Aldrin spent almost 22 hours on the moon. About 2.5 hours of this was spent outside the Eagle, collecting rock and soil samples, setting up experiments, and taking pictures.

What was it like?
Buzz Aldrin described the moon's surface as like nothing on Earth. He said it consisted of a fine, talcum-powder-like dust, strewn with pebbles and rocks.

Here comes Earth
Instead of the moon rising, the astronauts saw Earth rising over the moon's horizon – it looked four times bigger than the moon looks from Earth.

We have transport!
Three later Apollo missions each carried a small electric car, a lunar rover, which allowed the astronauts to explore away from the lander. These were left on the moon when the astronauts left.

Splashdown
The astronauts returned to Earth in the Apollo 11 command module. This fell through the atmosphere and landed in the Pacific Ocean. A ringed float helped to keep it stable.

How did they talk?
There's no air in space, so sound has nothing to travel through. Lunar astronauts use radio equipment in their helmets.

Why is there no blue sky on the moon?

One lunar rover reached a top speed of 22 km/h (13.5 mph).

The dish antennas allowed the astronauts to send pictures to Earth.

Neil Armstrong

Colour coding identifies each chapter at a glance.

weird or what?

Want to know something surprising? Then look at a "Weird or what?" tip.

Weird or what? are packed with extra weird or wonderful facts.